RUNTIME TOPOLOGY & CABLING

Continuum Runtime Diagram

Every physical component, cable, and network hop in the reference deployment — drawn from the actual configs in `image/buildroot/config-templates/`, not the abstract architecture.

Doc ARCH-003 **Companion to** ARCH-001 / ARCH-002

Cluster 3 control-plane + 3 compute, minimum fault-tolerant reference

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01 One node's wiring, in full

Every node in the cluster — control-plane or compute, since ARCH-002 §01 puts them on identical Pi 5 hardware — is wired identically to its own boot processor. This is that wiring, once, in detail; §02 shows how all six repeat it into one fabric.

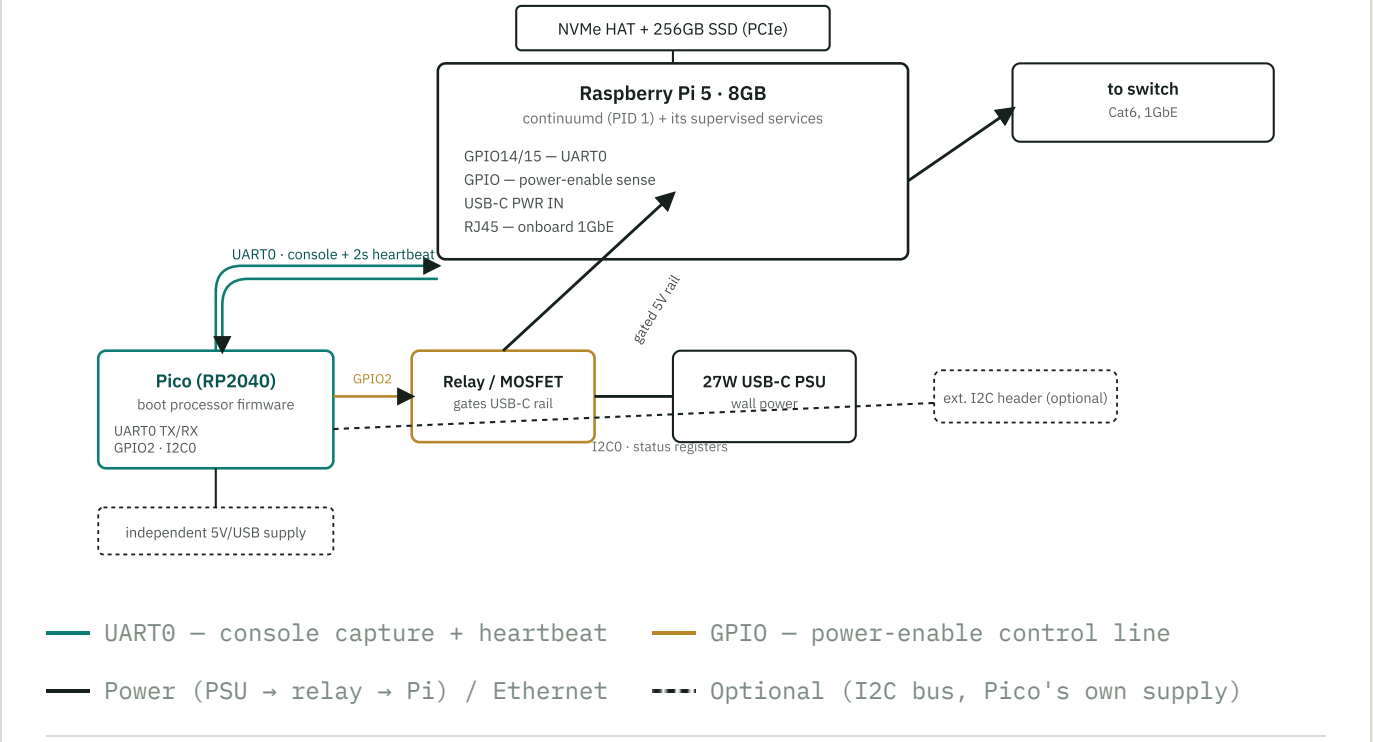


Fig. 1 — One node, fully wired. The relay sits between the wall PSU and the Pi's USB-C input specifically so the boot processor can cut power without being in the Pi's

own power path (firmware/boot_processor/src/main.c).

02 Cluster fabric — six boards, one switch

Figure 1's wiring repeats six times — 3 control-plane, 3 compute — with every node's Ethernet port landing on the same flat 1GbE segment. Nothing here is asymmetric at the network layer; role is entirely a software distinction (§03).

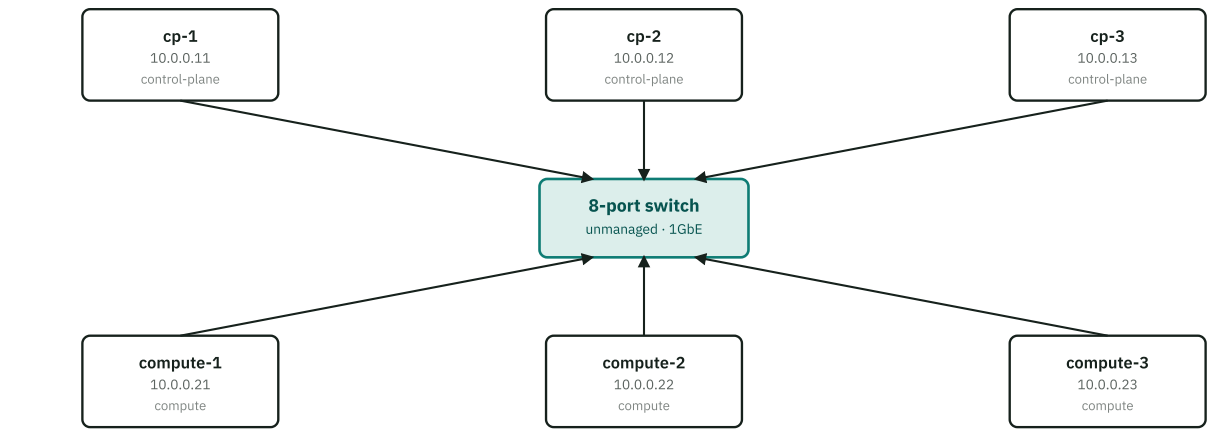


Fig. 2 — Flat, unmanaged 1GbE segment (ARCH-001 §07). IPs shown match image/buildroot/config-templates/'s example addressing; a real deployment can use DHCP reservations instead.

03 Software runtime — what actually talks to what

One control-plane node and one compute node, expanded to their supervised processes, with the **real ports and transports from the built code** — not the abstract architecture. Every arrow here is a config line in `image/buildroot/config-templates/`.

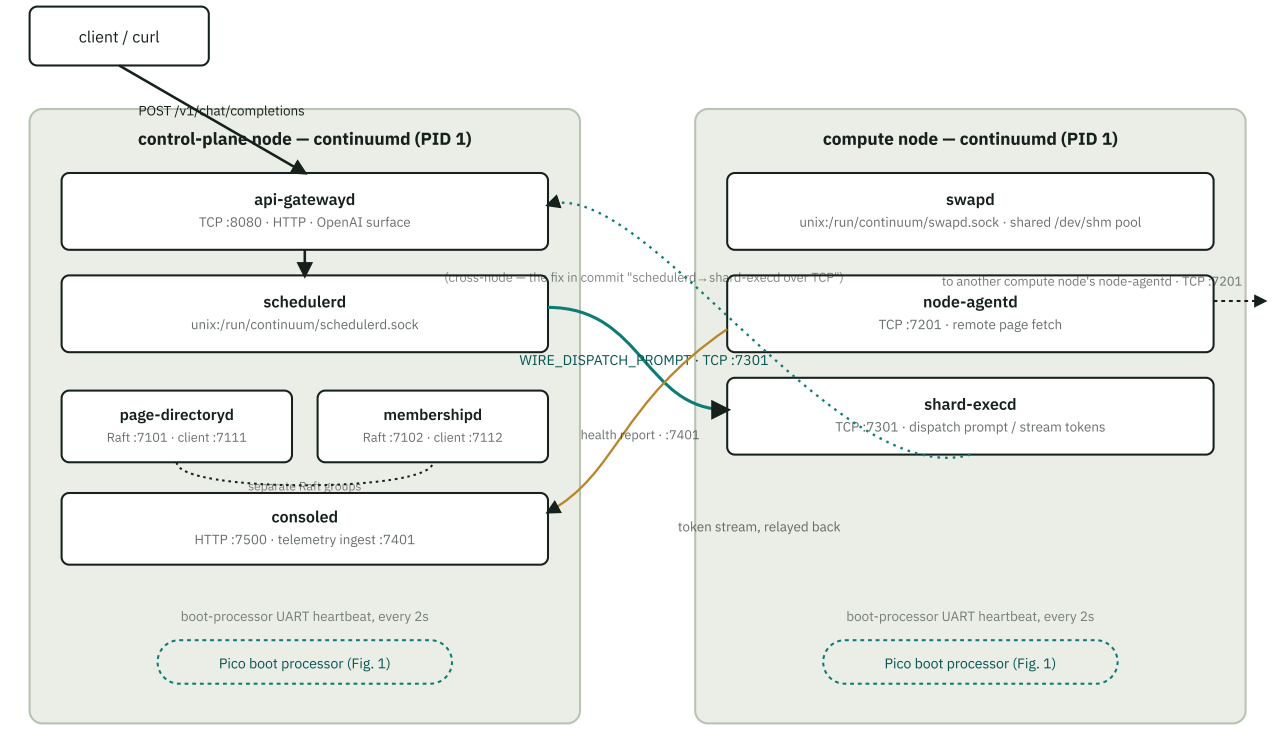


Fig. 3 — Every port here is a real default from the service's own *_config.c / *_config.h. page-directoryd and membershipd each run their own Raft group across all 3 control-plane replicas (only the local pair is drawn — see Fig. 2 for the full mesh).

Continuum Runtime Diagram — ARCH-003, companion to ARCH-001/ARCH-002 · drawn from the built repository, not the design doc.

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